Hints: Combinatorics – 4. Graphs

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Handshake: every edge contributes 1 to the degree of each of its two ends. For trees, first show that an acyclic graph with at least one edge has a leaf (take a longest path: its end has no other neighbour), and that deleting a leaf keeps a tree. Induction then gives $|E| = n - 1$ for trees, and the other characterisations follow by counting components: a forest with c components has $n - c$ edges.

Hint 2. Prüfer code: repeatedly delete the leaf with the smallest label and write down its neighbour, until two vertices remain. A vertex of degree d appears exactly $d-1$ times in the word, so the smallest label missing from the word is the first deleted leaf; this lets you rebuild the tree from the word step by step. Hence the map is a bijection onto all n^{n-2} words.

Hint 3. A connected graph with n vertices and n edges is not a tree, so it has a cycle; deleting an edge of that cycle leaves a connected graph with $n - 1$ edges, i.e. a tree, in which the cycle edge closes exactly one cycle. If there were two cycles, deleting an edge lying on only one of them would leave a cycle in a tree.

Hint 4. Counting: choose the cycle on $k \geqslant 3$ vertices in $\binom{n}{k}\frac{(k-1)!}{2}$ ways, then attach a rooted forest whose roots are the k cycle vertices. The number of forests on n labelled vertices with k given roots is $k\,n^{n-k-1}$ (generalised Cayley: glue the roots to a new vertex, or adapt the Prüfer code). The answer is

$$\sum_{k=3}^{n} \binom{n}{k} \frac{(k-1)!}{2}\, k\, n^{n-k-1} = \frac{1}{2} \sum_{k=3}^{n} \frac{n!}{(n-k)!}\, n^{n-k-1}.$$

Problem 2

Hint 1. Hall: necessity is clear. For sufficiency follow the sketch in Theorem 3. In the critical case $|N(S)| = |S|$, check Hall's condition for the graph between S and $N(S)$ (trivially inherited) and for the graph between $A \setminus S$ and $B \setminus N(S)$: if $T \subseteq A \setminus S$ had $|N(T) \setminus N(S)| < |T|$, then $S \cup T$ would violate the condition in the original graph.

Hint 2. k-regular: the $k|S|$ edges leaving S all land in $N(S)$, which receives at most $k|N(S)|$ edges, so $|N(S)| \geqslant |S|$. Remove a perfect matching; the rest is $(k-1)$-regular, and induct.

Hint 3. Minimum degree $\geqslant n/2$: if $|S| \leqslant n/2$, any vertex of S already has $\geqslant n/2 \geqslant |S|$ neighbours. If $|S| > n/2$, every $b \in B$ has $\geqslant n/2 > |A \setminus S|$ neighbours, so some neighbour lies in S; hence $N(S) = B$.

Hint 4. For the counterexample one edge too few is enough: $A = \{a_1, a_2\}$, $B = \{b_1, b_2\}$, both a_i joined only to b_1.

Problem 3

Hint 1. Work with one component at a time (n_i vertices, m_i edges, c_i cycles). Delete one edge from each of the c_i cycles. Since every cycle of the graph is one of these, the result has no cycle and is still connected, so $m_i - c_i = n_i - 1$.

Hint 2. The cycles are edge-disjoint and each has at least 3 edges, so $3c_i \leqslant m_i$. Combine: $m_i \leqslant n_i - 1 + m_i/3$, i.e. $2m_i \leqslant 3n_i - 3 < 3n_i$. Sum over components.

Problem 4

Hint 1. Count paths of length 2 ("cherries" $u-w-v$): there are $\sum_i \binom{a_i}{2} = \frac{n^2-n-2m}{2} = \binom{n}{2} - m$, which is exactly the number of non-adjacent pairs.

Hint 2. Every non-adjacent pair is the end pair of at least one cherry. Equality of the two counts therefore forces: no cherry has adjacent ends (no triangles), and every non-adjacent pair has exactly one common friend (so no 4-cycles either). Hence $k \geqslant 5$.

Hint 3. Every vertex has degree $\geqslant 2$: if u had a single friend w, then w would be the common friend of u and everybody else, i.e. a friend of everybody. So the graph has cycles. Take a shortest cycle; if its length were $\geqslant 6$, two of its vertices at distance 3 along it are at distance $\leqslant 2$ in the graph, and the short path together with the shorter arc gives a shorter cycle.

Hint 4. So $k = 5$, and it occurs: the pentagon C_5 satisfies all conditions ($\sum a_i^2 = 20 = 5^2 - 5$); so does the Petersen graph.

Problem 5

Hint 1. Induction on the number of edges: a tree has $f = 1$ and $e = v - 1$. Otherwise delete an edge lying on a cycle; it separates two different faces (Jordan curve theorem for the cycle), which merge, so e and f both drop by 1.

Hint 2. Count edge–face incidences: every face of a graph with $v \geqslant 3$ is bounded by at least 3 edge sides and every edge has two sides, so $3f \leqslant 2e$; substitute into Euler's formula to get $e \leqslant 3v - 6$, with equality iff every face is a triangle. Without triangles faces have $\geqslant 4$ sides, $4f \leqslant 2e$, and $e \leqslant 2v - 4$.

Hint 3. K_5: $10 > 3 \cdot 5 - 6$. $K_{3,3}$ is triangle-free and $9 > 2 \cdot 6 - 4$. Finally $\sum \deg v = 2e \leqslant 6v - 12 < 6v$, so some vertex has degree $\leqslant 5$.